

A Counterexample to Conjecture 000000000006

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Claude (Opus 5)[‡]

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Abstract

Conjecture 000000000006 asserts that a set $A \subseteq [N]$ in which the average of any two elements of the same parity is never prime satisfies $|A| \leq (1/3 + \varepsilon)N$, with $1/3$ attained by the multiples of three. We exhibit an admissible set of density $5/12$, refuting the upper bound. The conjecture's lower-bound construction is correct; only its extremality fails. Both halves of the refutation — admissibility and the density count — are formalized in Lean 4 against Mathlib.

1 Statement and reading

Conjecture 000000000006 concerns a set $A \subseteq \{1, \dots, N\}$ such that the average of any two elements of the same parity is not prime, and asserts

$$|A| \leq \left(\frac{1}{3} + \varepsilon\right) N \quad (1)$$

for every fixed $\varepsilon > 0$ and all sufficiently large N .

The two elements must be required to be *distinct*. Otherwise, taking $a = b = p$ for any prime $p \in A$ gives $(p + p)/2 = p$, so no set containing a prime is admissible at all, and the conjecture degenerates. We therefore read the hypothesis as

$$a, b \in A, \quad a \neq b, \quad a \equiv b \pmod{2} \implies \frac{a+b}{2} \text{ is not prime.} \quad (2)$$

Under this reading, (1) is false.

2 The counterexample

Definition 1. Let

$$A = \{n \geq 1 : 4 \mid n\} \cup \{n \geq 1 : n \equiv 3 \pmod{6}\}.$$

Equivalently $A = \{n \geq 1 : n \bmod 12 \in \{0, 3, 4, 8, 9\}\}$: the residues $0, 4, 8$ are precisely the even residues divisible by 4, and $3, 9$ are precisely the odd residues congruent to 3 modulo 6. The two parts are disjoint, the first consisting of even and the second of odd numbers.

Proposition 2. A satisfies (2).

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Proof. Let $a \neq b$ be elements of A of the same parity.

Both even. Then a and b lie in the first part, so $4 \mid a$ and $4 \mid b$, hence $a + b \equiv 0 \pmod{4}$ and

$$\frac{a+b}{2} \equiv 0 \pmod{2}.$$

The two smallest distinct positive multiples of 4 are 4 and 8, so $a + b \geq 12$ and $(a + b)/2 \geq 6$. Thus the average is an even integer greater than 2, and is not prime.

Both odd. Then $a \equiv b \equiv 3 \pmod{6}$, so $a + b \equiv 0 \pmod{6}$ and

$$\frac{a+b}{2} \equiv 0 \pmod{3}.$$

The two smallest distinct positive integers congruent to 3 modulo 6 are 3 and 9, so $a + b \geq 12$ and $(a + b)/2 \geq 6$. Thus the average is a multiple of 3 greater than 3, and is not prime.

Pairs of opposite parity are unconstrained by (2). $\square$

The two divisibility steps are exactly where the congruences are used, and neither follows from parity and distinctness alone: 2 and 4 are distinct even numbers with average 3, and 1 and 3 are distinct odd numbers with average 2. Likewise the bounds $(a + b)/2 > 2$ and $(a + b)/2 > 3$ come from the smallest admissible pairs, not from distinctness.

3 The density of A

Proposition 3. *For every $N \geq 1$,*

$$|A \cap [1, N]| = \left\lfloor \frac{N}{4} \right\rfloor + \left\lfloor \frac{N+3}{6} \right\rfloor = \frac{5}{12}N + O(1).$$

In particular $12|A \cap [1, N]| + 55 \geq 5N$ for every N .

Proof. The two parts of A are disjoint. The multiples of 4 in $[1, N]$ number $\lfloor N/4 \rfloor$; the integers $n \equiv 3 \pmod{6}$ in $[1, N]$ are 3, 9, ..., and number $\lfloor (N+3)/6 \rfloor$. Their sum is $N/4 + N/6 + O(1) = \frac{5}{12}N + O(1)$.

For the explicit form, write $N = 12q + r$ with $0 \leq r < 12$. Each block $(12j, 12j + 12]$ contains exactly five elements of A , namely $12j + 3, 12j + 4, 12j + 8, 12j + 9, 12j + 12$, so $|A \cap [1, 12q]| = 5q$. Since $12q \geq N - 11$, we get $12|A \cap [1, N]| \geq 60q \geq 5(N - 11) = 5N - 55$. $\square$

4 The conjecture is false

Theorem 4. *A satisfies (2), and for every $\varepsilon < 1/12$ the bound (1) fails for all sufficiently large N . Concretely, with $\varepsilon = 1/24$,*

$$|A \cap [1, N]| > \left(\frac{1}{3} + \frac{1}{24} \right) N = \frac{3}{8}N \quad \text{for every } N \geq 111.$$

Proof. Admissibility is Proposition 2. By Proposition 3, $24|A \cap [1, N]| \geq 10N - 110$, which exceeds $9N$ as soon as $N > 110$. Since $5/12 > 1/3 + \varepsilon$ for every $\varepsilon < 1/12$, the same argument applies to any such ε . $\square$

We note that the conjecture's lower-bound construction is sound: for distinct positive multiples a, b of 3 of the same parity, $(a + b)/2$ is a multiple of 3 exceeding 3, hence composite. The multiples of three are admissible and have density $1/3$. What fails is only the claim that this is extremal.

5 Remark: 5/12 appears to be optimal

Write $M(N) = \max |A|$ over admissible $A \subseteq [N]$. Condition (2) never relates an even element to an odd one, so the problem splits and $M(N) = M_{\text{ev}}(N) + M_{\text{od}}(N)$.

For the even part, put $a = 2x$; the condition becomes “ $x + y$ is not prime” for distinct $x, y \leq \lfloor N/2 \rfloor$. An odd prime is a sum of two integers of opposite parity, and $x + y = 2$ forces $x = y = 1$; hence every forbidden pair joins an even x to an odd x , the conflict graph is bipartite, and its maximum independent set is computable exactly by König’s theorem. For the odd part, put $a = 2x + 1$; the condition becomes “ $x + y + 1$ is not prime”, and we computed the maximum by branch and bound.

N	720	1440	2880	5760	11520
$M_{\text{ev}}(N)$	180	360	720	1440	2880
$\lfloor N/4 \rfloor$	180	360	720	1440	2880

N	480	600	720
$M_{\text{od}}(N)$	80	100	120
$\lfloor N/6 \rfloor$	80	100	120

In every case computed, $M_{\text{ev}}(N) = \lfloor N/4 \rfloor$ and $M_{\text{od}}(N) = \lfloor N/6 \rfloor$, so $M(N) = \frac{5}{12}N + O(1)$ and the set A above is extremal. We have not proved this, and record it only as computational evidence. It suggests the repaired statement:

If $A \subseteq [N]$ has no two distinct elements of the same parity whose average is prime, then $|A| \leq (5/12 + \varepsilon)N$, and 5/12 is attained by $\{n : 4 \mid n\} \cup \{n : n \equiv 3 \pmod{6}\}$.

6 Formal verification

The accompanying Lean 4 project formalizes both halves of the refutation against Mathlib, using Mathlib’s `Nat.Prime` throughout rather than a locally defined notion of primality.

- `inA_admissible` — Proposition 2: for distinct positive $a, b \in A$ of equal parity, $\neg \text{Nat.Prime}((a + b)/2)$. The underlying case analysis `average_obstruction` covers all twenty-five combinations of the five selected residue classes.
- `countA_lower` — Proposition 3 in the explicit form $5N \leq 12 \text{countA } N + 55$, where $\text{countA } N = |A \cap [1, N]|$. The lower bound is obtained from an explicit order-preserving enumeration `enumA` of A , shown injective and to land in $[1, 12k]$ on $\{0, \dots, 5k - 1\}$.
- `conjecture_00000000006_false` — the two combined: admissibility together with $9N < 24 \text{countA } N$ for all $N \geq 111$.
- `multiples_of_three_admissible` — the conjecture’s own lower-bound construction, for completeness.

The project builds with `lake build on leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000006_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.